

Examples from “Debug Javascript”

```
function clickedTreatButton() {  
  // Checks if Raya is too (hungry, tired, or sad) to play  
  if (!canPerformAction("eat a treat")) return;  
  
  // Setting a weight limit before Raya gets overfed  
  const weightLimit = 40;  
  
  // Normal boost when pet is within a healthy weight  
  if (pet_info.weight < weightLimit) {  
  
    pet_info.hapiness += 5; // <----- BUG: TYPO - happiness missing 'p'  
    pet_info.weight += 5;  
    pet_info.energy += 3;  
  
    statusUpdate(`${pet_info.name} loved the treat! 🍪`);  
  }  
}
```

BUG: clickedTreatButton() will not properly increase happiness b/c of typo (missing 'p')

Project 2 - Giga Pet



Select your pet

Raya

Spotty

Casper

Ziggy

Name: **Raya**

Weight: **20 pounds**

Happiness: **10 tail wags (per min)**

Energy: **8** ⚡

Treat

Play

Exercise

Sleep

Paused in debugger

Elements Console Sources Network Performance Memory Application

jquery-2.2.1.min.js script.js X

```
97
98 function clickedTreatButton() {
99   debugger; // <---- use debugger so that dev tools jumps from jQuery file
100   // Checks if Raya is too (hungry, tired, or sad) to play
101   if (!canPerformAction("eat a treat")) return;
102
103   // Setting a weight limit before Raya gets overfed
104   const weightLimit = 40;
105
106   // Normal boost when pet is within a healthy weight
107   if (pet_info.weight < weightLimit) {
108
109     pet_info.hapiness += 5; // <----- BUG: TYPO - happiness missing 'p'
110     pet_info.weight += 5;
111     pet_info.energy += 3;
112
113     statusUpdate(`${pet_info.name} loved the treat! 🍪`);
114   }
115   // If the pet is overfed, eating treats makes them unhappy
116   else {
117
118     pet_info.weight += 3;
119     pet_info.happiness -= 4;
120     pet_info.energy -= 2;
121
122     statusUpdate(`${pet_info.name} is too full and feels sick... 🤢`);
123
124     $(".pet-image").addClass("overfed");
125     setTimeout(() => $(".pet-image").removeClass("overfed"), 800);
126
127   }
128   checkAndUpdatePetInfoInHtml();

```

clickedTreatButton script.js:99

dispatch jquery-2.2.1.min.js:3

r.handle jquery-2.2.1.min.js:3

XHR/fetch Breakpoints

DOM Breakpoints

Global Listeners

Event Listener Breakpoints

Filter

☐ Ad Auction Worklet

☐ Animation

☐ Canvas

☐ Clipboard

☐ Control

☐ Device

☐ DOM Mutation

☐ Drag / drop

☐ Geolocation

☐ Keyboard

☐ Load

☐ Media

☒ Mouse

☐ auxclick

☒ click

☐ dblclick

☐ mousedown

☐ mouseup

☐ mouseover

☐ mousemove

☐ mouseout

☐ mouseenter

Line 99, Column 7

Coverage: n/a

Relevant data is sent to Google

Paused line of code

Project 2 - Giga Pet



Select your pet

Raya

Spotty

Casper

Ziggy

Name: **Raya**

Weight: **20 pounds**

Happiness: **10 tail wags (per min)**

Energy: **8** ⚡

Treat

Play

Exercise

Sleep

Paused in debugger

Elements Console Sources Network Performance Memory Application

jquery-2.2.1.min.js script.js X

```
97
98 function clickedTreatButton() {
99   debugger; // <---- use debugger so that dev tools jumps from jQuery file
100   // Checks if Raya is too (hungry, tired, or sad) to play
101   if (!canPerformAction("eat a treat")) return;
102
103   // Setting a weight limit before Raya gets overfed
104   const weightLimit = 40;
105
106   // Normal boost when pet is within a healthy weight
107   if (pet_info.weight < weightLimit) {
108
109     pet_info.hapiness += 5; // <----- BUG: TYPO - happiness missing 'p'
110     pet_info.weight += 5;
111     pet_info.energy += 3;
112
113     statusUpdate(`${pet_info.name} loved the treat! 🍪`);
114   }
115   // If the pet is overfed, eating treats makes them unhappy
116   else {
117
118     pet_info.weight += 3;
119     pet_info.happiness -= 4;
120     pet_info.energy -= 2;
121
122     statusUpdate(`${pet_info.name} is too full and feels sick... 🤢`);
123
124     $(".pet-image").addClass("overfed");
125     setTimeout(() => $(".pet-image").removeClass("overfed"), 800);
126
127   }
128   checkAndUpdatePetInfoInHtml();

```

clickedTreatButton script.js:101

dispatch jquery-2.2.1.min.js:3

r.handle jquery-2.2.1.min.js:3

XHR/fetch Breakpoints

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Global Listeners

Event Listener Breakpoints

Filter

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☐ Keyboard

☐ Load

☐ Media

☒ Mouse

☐ auxclick

☒ click

☐ dblclick

☐ mousedown

☐ mouseup

☐ mouseover

☐ mousemove

☐ mouseout

☐ mouseenter

Line 101, Column 7

Coverage: n/a

Relevant data is sent to Google

Step through code

Project 2 - Giga Pet



Select your pet

Raya

Spotty

Casper

Ziggy

Name: **Raya**

Weight: **20 pounds**

Happiness: **10 tail wags (per min)**

Energy: **8** ⚡

Treat

Play

Exercise

Sleep

Paused in debugger

Elements Console Sources Network Performance Memory Application

jquery-2.2.1.min.js script.js X

```
94 }
95
96 // ----- ACTIONS -----
97
98 function clickedTreatButton() {
99   debugger; // <---- use debugger so that dev tools jumps from jQuery file
100   // Checks if Raya is too (hungry, tired, or sad) to play
101   if (!canPerformAction("eat a treat")) return;
102
103   // Setting a weight limit before Raya gets overfed
104   const weightLimit = 40; weightLimit = 40
105
106   // Normal boost when pet is within a healthy weight
107   if (pet_info.weight < weightLimit) { weightLimit = 40
108
109   pet_info.hapiness += 5; // <----- BUG: TYPO - happiness missing 'p'
110   pet_info.weight += 5;
111   pet_info.energy += 3;
112
113   statusUpdate(`${pet_info.name} loved the treat! 🍪`);
114 }
115 // If the pet is overfed, eating treats makes them unhappy
116 else {
117
118   pet_info.weight += 3;
119   pet_info.happiness -= 4;
120   pet_info.energy -= 2;
121
122   statusUpdate(`${pet_info.name} is too full and feels sick... 🤢`);
123
124   $(".pet-image").addClass("overfed");
125   setTimeout(() => $(".pet-image").removeClass("overfed"), 800);
```

clickedTreatButton script.js:109

dispatch jquery-2.2.1.min.js:3

r.handle jquery-2.2.1.min.js:3

XHR/fetch Breakpoints

DOM Breakpoints

Global Listeners

Event Listener Breakpoints

Filter

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☐ Keyboard

☐ Load

☐ Media

☒ Mouse

☐ auxclick

☒ click

☐ dblclick

☐ mousedown

☐ mouseup

☐ mouseover

☐ mousemove

☐ mouseout

☐ mouseenter

Line 109, Column 9 Coverage: n/a Relevant data is sent to Google

Using Breakpoints

Project 2 - Giga Pet



Select your pet

- Raya
- Spotty
- Casper
- Ziggy

Name: **Raya**
Weight: **20 pounds**
Happiness: **10 tail wags (per min)**
Energy: **8** ⚡

- Treat
- Play
- Exercise
- Sleep

ElementsSourcesConsoleNetworkPerformanceMemoryApplication>>

topFilterDefault levelsNo Issues

```
> pet_info.hapiness += 5;
< NaN
> pet_info.happiness += 5;
< 15
>
```

☐ Relevant data is sent to Google

Identifying the issue of happiness not increasing is due to a typo (“hapiness” is an undefined var)

Project 2 - Giga Pet



Select your pet

- Raya
- Spotty
- Casper
- Ziggy

Name: **Raya**
Weight: **25 pounds**
Happiness: **15 tail wags (per min)**
Energy: **11 ⚡**

- Treat
- Play
- Exercise
- Sleep

Elements Sources Console Network Performance Memory Application

jquery-2.2.1.min.js script.js

```
94 }
95
96 // ----- ACTIONS -----
97
98 function clickedTreatButton() {
99   debugger; // <---- use debugger so that dev tools jumps from jQuery file
100   // Checks if Raya is too (hungry, tired, or sad) to play
101   if (!canPerformAction("eat a treat")) return;
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103   // Setting a weight limit before Raya gets overfed
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105
106   // Normal boost when pet is within a healthy weight
107   if (pet_info.weight < weightLimit) {
108
109     pet_info.happiness += 5; // <----- BUG: TYPO - happiness missing 'p'
110     pet_info.weight += 5;
111     pet_info.energy += 3;
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113     statusUpdate(`${pet_info.name} loved the treat! 🍪`);
114   }
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118     pet_info.weight += 3;
119     pet_info.happiness -= 4;
120     pet_info.energy -= 2;
121
122     statusUpdate(`${pet_info.name} is too full and feels sick... 🤢`);
123
124     $(".pet-image").addClass("overfed");
125     setTimeout(() => $(".pet-image").removeClass("overfed"), 800);
126   }
127 }
```

Watch

No watch expressions

Breakpoints

☐ Pause on uncaught exceptions

☐ Pause on caught exceptions

script.js

☒ pet_info.happiness... 109

Scope

Call Stack

Not paused

XHR/fetch Breakpoints

DOM Breakpoints

Global Listeners

Event Listener Breakpoints

Filter

☐ Ad Auction Worklet

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☐ Canvas

☐ Clipboard

☐ Control

☐ Device

☐ DOM Mutation

☐ Drag / drop

☐ Geolocation

☐ Keyboard

☐ Load

☐ Media

☒ Mouse

Line 109, Column 38

Coverage: n/a

Relevant data is sent to Google

Applying fix: Happiness properly increases by 5

Examples from “View DOM nodes”

Project 2 - Giga Pet



Select your pet

- Raya
- Spotty
- Casper
- Ziggy

Name: **Raya**
Weight: **20 pounds**
Happiness: **10 tail wags (per min)**
Energy: **8** ⚡

- Treat
- Play
- Exercise
- Sleep

ElementsSourcesConsoleNetworkPerformanceMemoryApplication

<!DOCTYPE html>
<html>
 <head>
 <body>
 <header>
 <main>
 <section class="pet-image-container">
 <section class="pet-switcher">
 <section class="dashboard">
 <div>Select your pet</div>
 <div class="pet-switcher">
 <div>
 <div>
 <div>
 <div>
 <div class="button-container">
 <button class="treat-button">Treat
 <button class="play-button">Play
 <button class="exercise-button">Exercise
 <button class="sleep-button">Sleep
 </div>
 </section>
 <section class="status-box">
 </main>
 <script src="https://code.jquery.com/jquery-2.2.1.min.js" integrity="sha256-gvQgAFzTH6trSrAwloH1iPo9Xc96QxSZ3feW6kem+000=" crossorigin="anonymous"></script>
 </body>
 </html>

htmlbodymainsection.dashboarddiv.button-containerbutton.treat-button

StylesComputedLayoutEvent Listeners

Filter: :hov .cls +, , ,

element.style {
}

.button-container button { style.css:70
 width: 100px;
 margin-right: 10px;
 text-align: center;
 display: inline-block;
 float: left;
 font-size: 15px;
 background-color: #1e2835;
 border-color: #404853;
 cursor: pointer;
 background-image: linear-gradient(to bottom, rgba(246, 246, 246, .1) 0%, rgba(30, 40, 53, 0) 66%);
 color: #fafafa;
}

* { style.css:2
 box-sizing: border-box;
}

button { user agent stylesheet
 appearance: auto;
 font-style: ;
 font-variant-ligatures: ;
 font-variant-caps: ;
 font-variant-numeric: ;
 font-variant-east-asian: ;
 font-variant-alternates: ;
 font-variant-position: ;
 font-variant-emoji: ;
 font-weight: ;
 font-stretch: ;
 font-size: ;
 font-family: ;
 font-optical-sizing: ;
 font-size-adjust: ;
 font-kerning: ;

Viewing DOM

Project 2 - Giga Pet



Select your pet

- Raya
- Spotty
- Casper
- Ziggy

Name: **Raya**
Weight: **20 pounds**
Happiness: **10 tail wags (per min)**
Energy: **8 ⚡**

- Snack
- Play
- Exercise
- Sleep

ElementsSourcesConsoleNetworkPerformanceMemoryApplication

<!DOCTYPE html>
<html>
 <head>
 <body>
 <header>
 <main>
 <section class="pet-image-container">
 <section class="pet-switcher">
 <section class="dashboard">
 <div>Select your pet</div>
 <div class="pet-switcher">
 <div>
 <div>
 <div>
 <div>
 "Energy: "
 == \$0
 8
 " ⚡ "

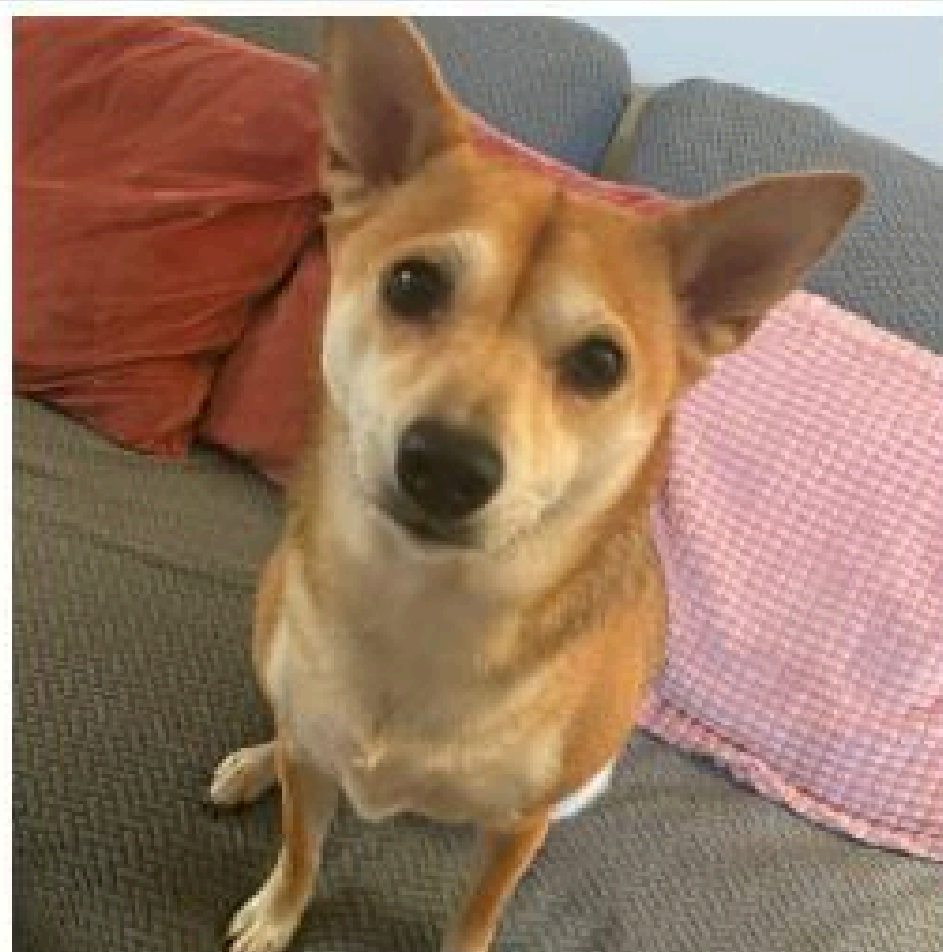
 </div>
 <div class="button-container">
 <button class="treat-button">Snack</button>
 <button class="play-button">Play</button>
 <button class="exercise-button">Exercise</button>
 <button class="sleep-button">Sleep</button>
 </div>
 </section>
 <section class="status-box">
 </main>
 <script src="https://code.jquery.com/jquery-2.2.1.min.js" integrity="sha256-gvQgAFzTH6trSrWoH1iPo9Xc96QxSZ3feW6kem+000=" crossorigin="anonymous"></script>
 </body>
</html>

StylesComputedLayoutEvent Listeners

Filter: :hov .cls +, , ,
element.style {
}
* {
 box-sizing: border-box;
}
strong {
 font-weight: bolder;
}
Inherited from body
body {
 font-family: helvetica, arial, sans-serif;
 margin: 25px;
}

margin 0
border 0
padding 0
auto x auto
0 0 0 0 0 0 0 0

Editing Content: Changing Treat to Snack



Select your pet

Raya

Spotty

Casper

Ziggy

Name: **Raya**

Weight: **20 pounds**

Happiness: **10 tail wags (per min)**

Energy:

8



Snack

Play

Exercise

Sleep

ElementsSourcesConsoleNetworkPerformanceMemoryApplication

<!DOCTYPE html>
<html> scroll
 <head> </head>
 <body>
 <header> </header>
 <main>
 <section class="pet-image-container"> </section>
 <section class="pet-switcher"> </section>
 <section class="dashboard">
 <div>Select your pet</div>
 <div class="pet-switcher"> </div>
 <div> </div>
 <div> </div>
 <div> </div>
 <div>
 "Energy: "
 <h1>
 8
 ⚡
 </h1>
 </div>
 <div class="button-container">
 <button class="treat-button">Snack</button>
 <button class="play-button">Play</button>
 <button class="exercise-button">Exercise</button>
 <button class="sleep-button">Sleep</button>
 </div>
 </section>
 <section class="status-box"> == \$0
 </main>
 <script src="https://code.jquery.com/jquery-2.2.1.min.js" integrity="sha256-gvQzAFzTH6trSrAwMoH1iPo9Xc96QxSZ3feW6kem+000=" crossorigin="anonymous"></script>
 </body>
</html>
html body main section.status-box

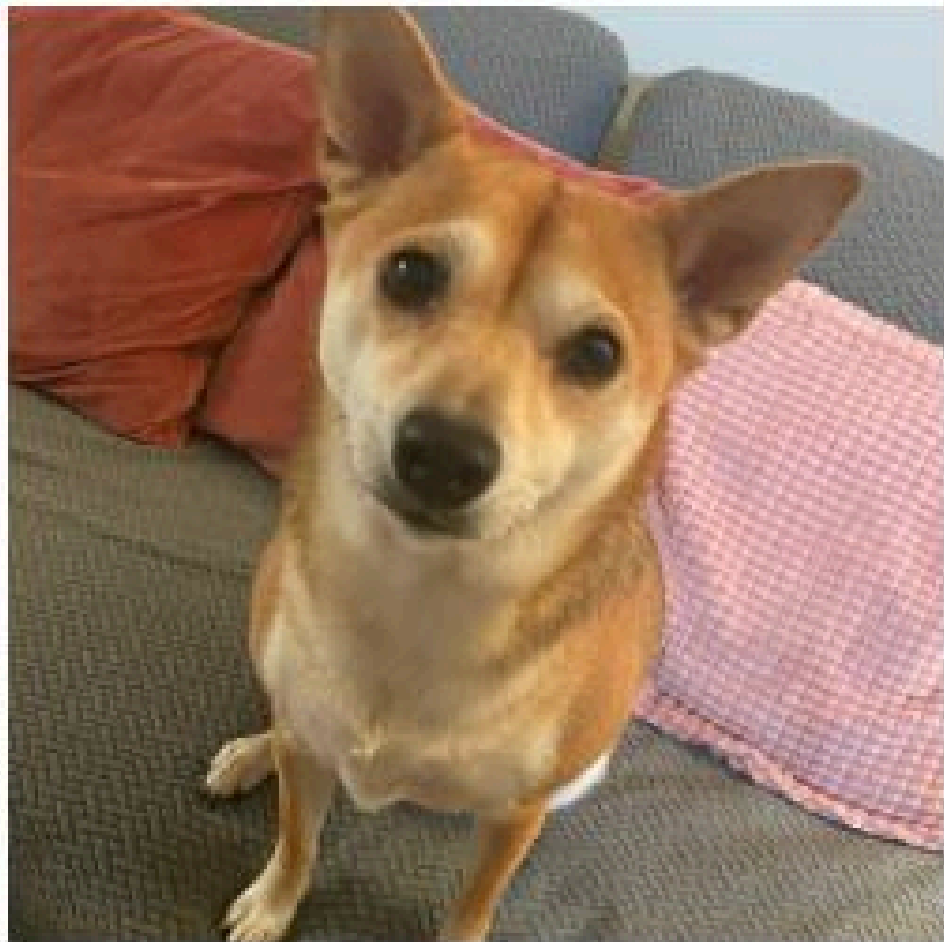
StylesComputedLayoutEvent Listeners

Filter: :hov .cls +, , ,
element.style {
}
.status-box { style.css:100
 clear: both;
 text-align: center;
}
* { style.css:2
 box-sizing: border-box;
}
section { user agent stylesheet
 display: block;
 unicode-bidi: isolate;
}
Inherited from body
body { style.css:6
 font-family: helvetica, arial, sans-serif;
 margin: 25px;
}

margin 0
border 0
padding 0
486x60
0 0 0 0 0 0 0 0

Editing Attribute: Changing to <h1>

Project 2 - Giga Pet



Select your pet

- Raya
- Spotty
- Casper
- Ziggy

Name: **Raya**
Weight: **20 pounds**
Happiness: **10 tail wags (per min)**

Energy: 8 

- Snack
- Play
- Exercise
- Sleep

ElementsSourcesConsoleNetworkPerformanceMemoryApplication

<!DOCTYPE html>
<html>
 <head>
 <body>
 <header>
 <main>
 <section class="pet-image-container">
 <section class="pet-switcher">
 <section class="dashboard">
 <div>Select your pet</div>
 <div class="pet-switcher">
 <div>
 <div>
 <div>
 <div id="Green">
 "Energy: "

 <div class="button-container">
 <button class="treat-button">Snack</button>
 <button class="play-button">Play</button>
 <button class="exercise-button">Exercise</button>
 <button class="sleep-button">Sleep</button>
 </div>
 </section>
 <section class="status-box">
 </main>
 <script src="https://code.jquery.com/jquery-2.2.1.min.js" integrity="sha256-gvQgAFzTH6trSrAw0H1iPo9Xc96QxS73few6kem+000=" crossorigin="anonymous"></script>
 </body>
</html>

StylesComputedLayoutEvent Listeners

Filter: :hov .cls +
element.style {
}
div#Green {
 background-color: limegreen;
}
.dashboard div {
 display: block;
}
* {
 box-sizing: border-box;
}
div {
 display: block;
 unicode-bidi: isolate;
}
Inherited from body
body {
 font-family: helvetica, arial, sans-serif;
 margin: 25px;
}

margin 0
border 0
padding 0
440x21
0 0 0 0 0 0 0 0

Adding an Attribute: Giving the “Energy” div an ID that applies a green background

Project 2 - Giga Pet



Select your pet

- Raya
- Spotty
- Casper
- Ziggy

Name: **Raya**
Weight: **20 pounds**
Happiness: **10 tail wags (per min)**
Energy: **8** ⚡

- Treat
- Play
- Exercise
- Sleep

ElementsSourcesConsoleNetworkPerformanceMemoryApplication

<!DOCTYPE html>
<html>
 <head>
 <body>
 <header>
 <main>
 <section class="pet-image-container">
 <section class="pet-switcher">
 <section class="dashboard">
 <div>
 <h3>Select your pet</h3>
 </div>
 <div class="pet-switcher">
 <div>
 <div>
 <div>
 <div>
 "Energy: "

 </div>
 <div class="button-container">
 </div>
 </section>
 <section class="status-box">
 </main>
 <script src="https://code.jquery.com/jquery-2.2.1.min.js" integrity="sha256-gvQgAFzTH6trSrAwof1iPo9Xc96QxSZ3feW6kem+000=" crossorigin="anonymous"></script>
 </body>
</html>

htmlbodymainsection.dashboarddivh3

StylesComputedLayoutEvent Listeners

Filter: :hov .cls +, , ,

element.style {
}
* {
 box-sizing: border-box;
}
h3 {
 display: block;
 font-size: 1.17em;
 margin-block-start: 1em;
 margin-block-end: 1em;
 margin-inline-start: 0px;
 margin-inline-end: 0px;
 font-weight: bold;
 unicode-bidi: isolate;
}
Inherited from body
body {
 font-family: helvetica, arial, sans-serif;
 margin: 25px;
}

margin18.720
border0
padding0
440x22
00000000

Using edit as HTML to insert the <h3> tags around "Select your pet"



Raya Spotty Casper Ziggy

Select your pet

Name: **Raya**
Weight: **20 pounds**
Happiness: **10 tail wags (per min)**
Energy: **8** ⚡

Treat Play Exercise Sleep

ElementsSourcesConsoleNetworkPerformanceMemoryApplication>>

<!DOCTYPE html>
<html> scroll
 <head> ... </head>
 <body>
 <header> ... </header>
 <main>
 <section class="pet-image-container"> ... </section>
 <section class="pet-switcher"> </section>
 <section class="dashboard">
 <div class="pet-switcher"> ... </div>
 ...
 <div> == \$0
 <h3>Select your pet</h3>
 </div>
 <div> ... </div>
 <div> ... </div>
 <div> ... </div>
 <div>
 "Energy: "
 ...
 </div>
 <div class="button-container"> ... </div>
 </section>
 <section class="status-box"> ... </section>
 </main>
 <script src="https://code.jquery.com/jquery-2.2.1.min.js" integrity="sha256-gvQgAFzTH6trSrAwWoH1iPo9Xc96QxSZ3feW6kem+000=" crossorigin="a" anonymous"></script>
 </body>
</html>

StylesComputedLayoutEvent Listeners>>

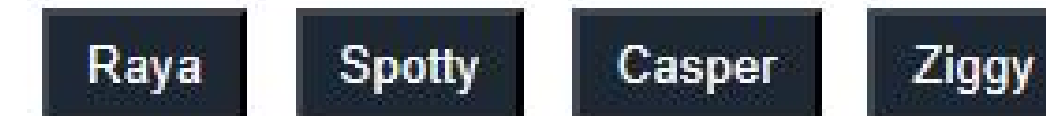
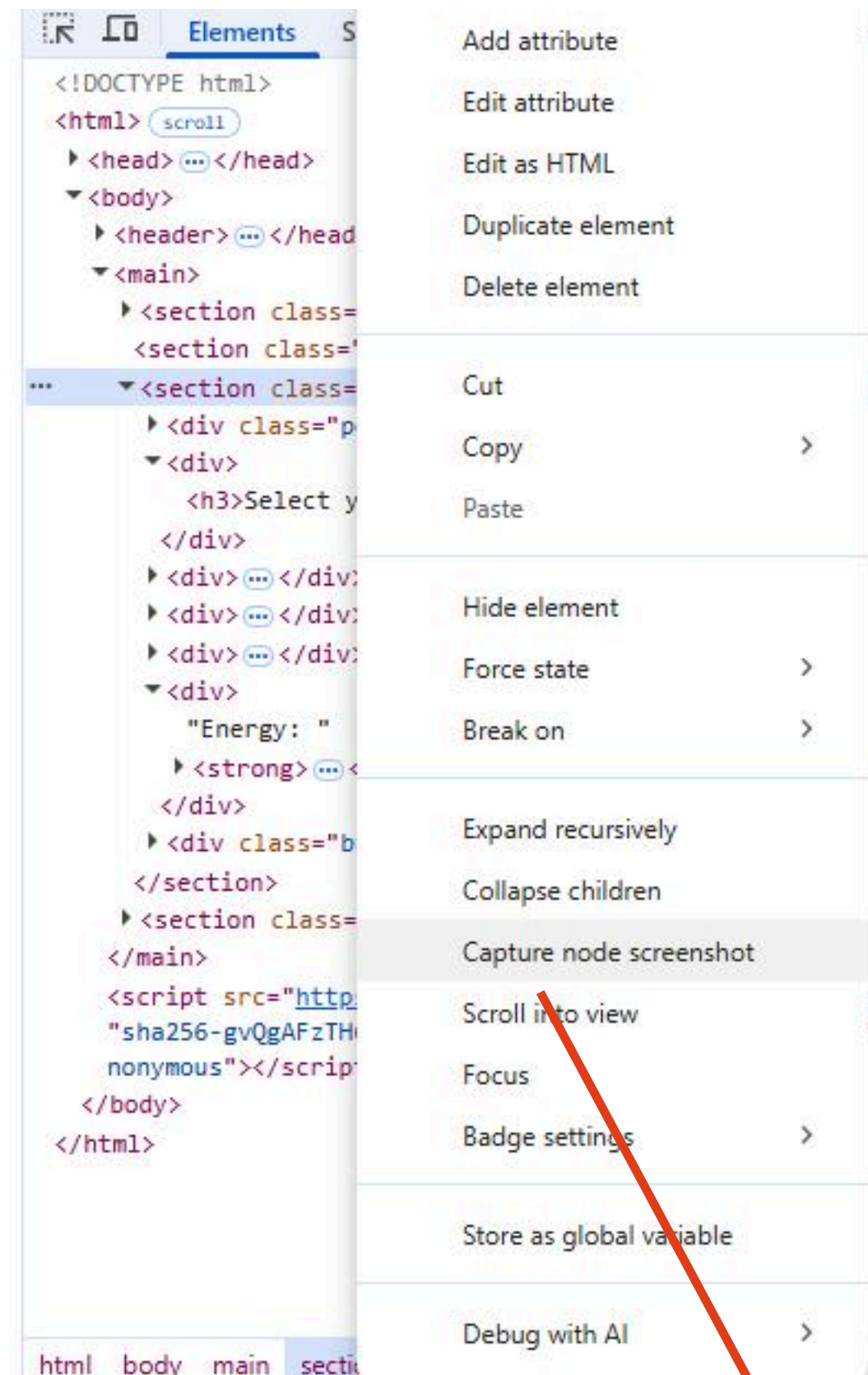
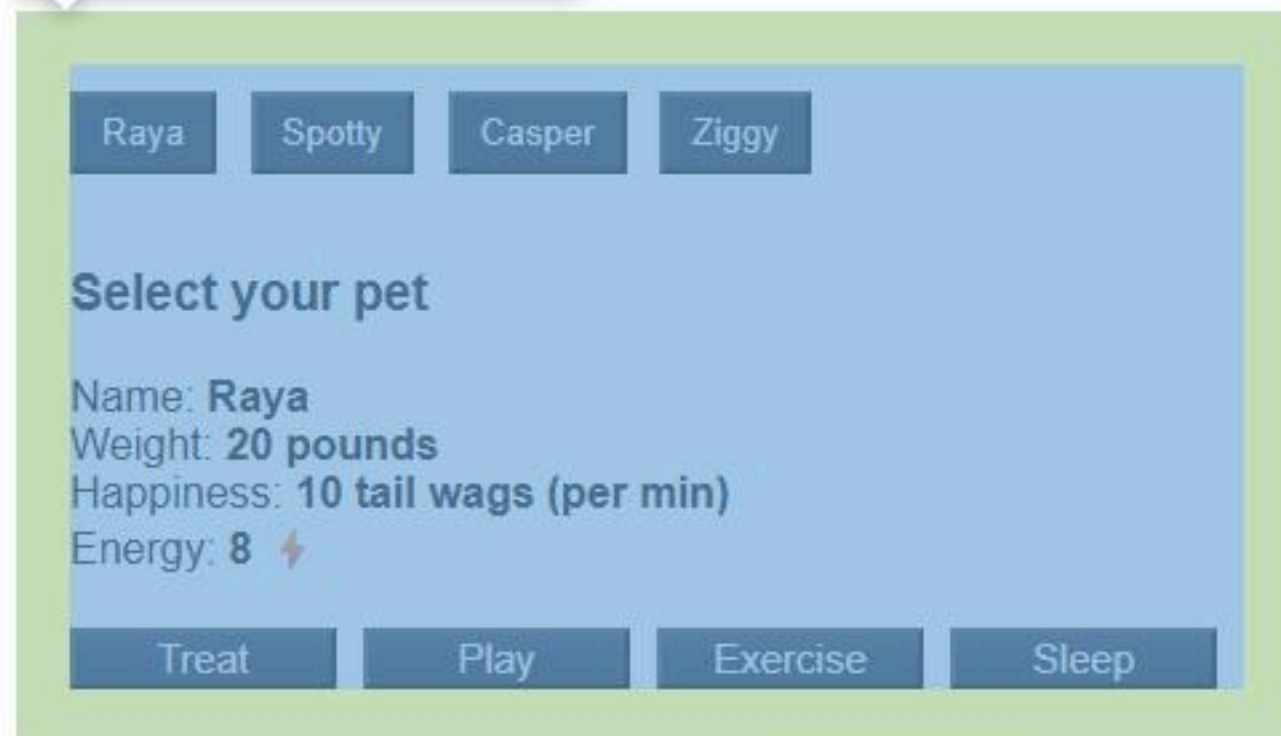
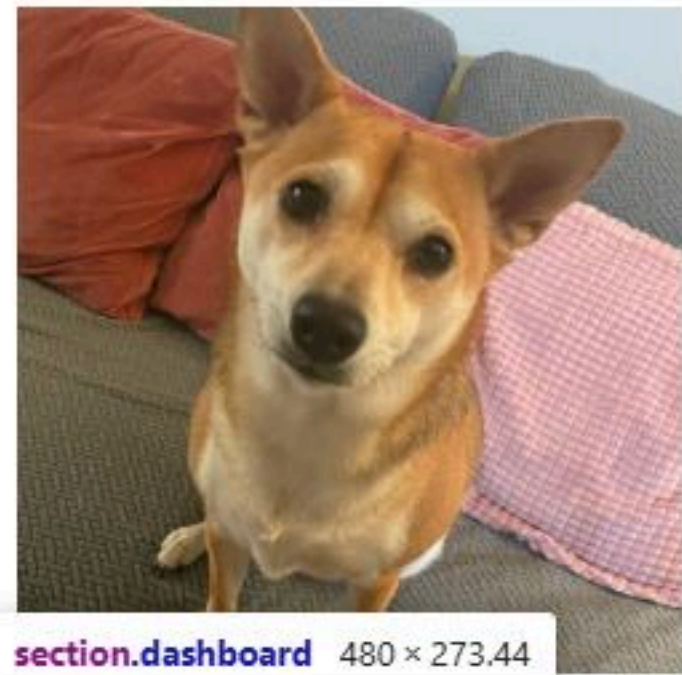
Filter: :hov .cls +, ->

element.style {
}
.dashboard div { style.css:52
 display: block;
}
* { style.css:2
 box-sizing: border-box;
}
div { user agent stylesheet
 display: block;
 unicode-bidi: isolate;
}
Inherited from body
body { style.css:6
 font-family: helvetica, arial, sans-serif;
 margin: 25px;
}

margin 0
border 0
padding 0
440x22
0 0 0 0 0 0 0 0

html body main section.dashboard div

Dragging the "Select your pet" div underneath the buttons



Select your pet

Name: **Raya**
Weight: **20 pounds**
Happiness: **10 tail wags (per min)**
Energy: **8 ⚡**



Using "Capture node screenshot"